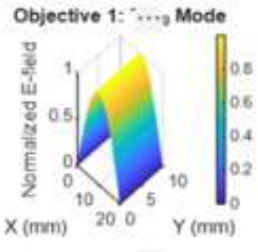
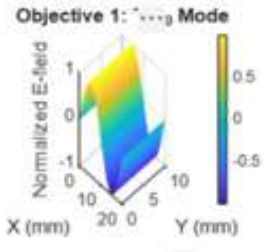
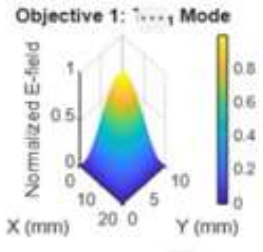
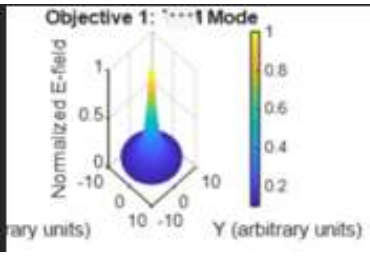
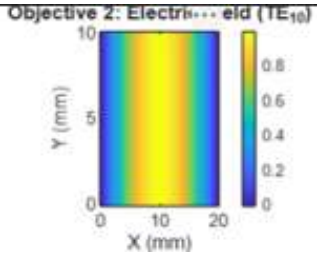
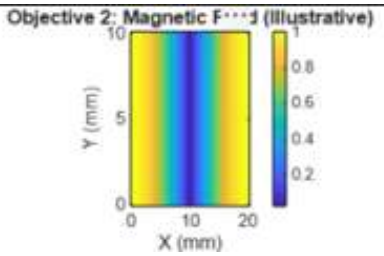

Objective 1: TM0 Mode
Normalized E-field
X (mm)
Y (mm)

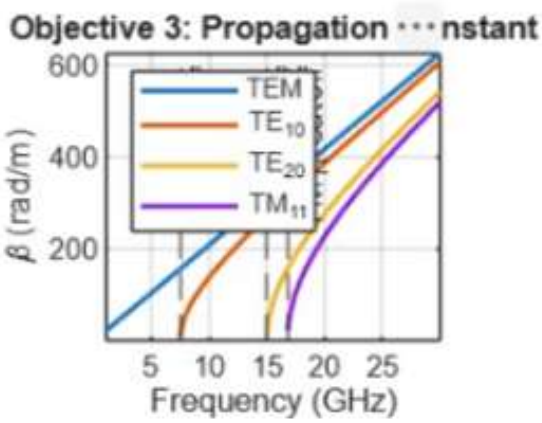

Objective 1: TM0 Mode
Normalized E-field
X (mm)
Y (mm)

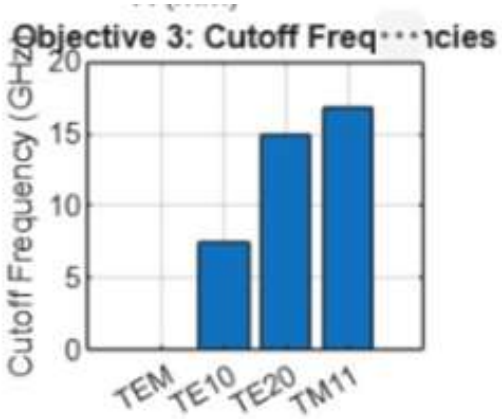

Objective 1: TM1 Mode
Normalized E-field
X (mm)
Y (mm)

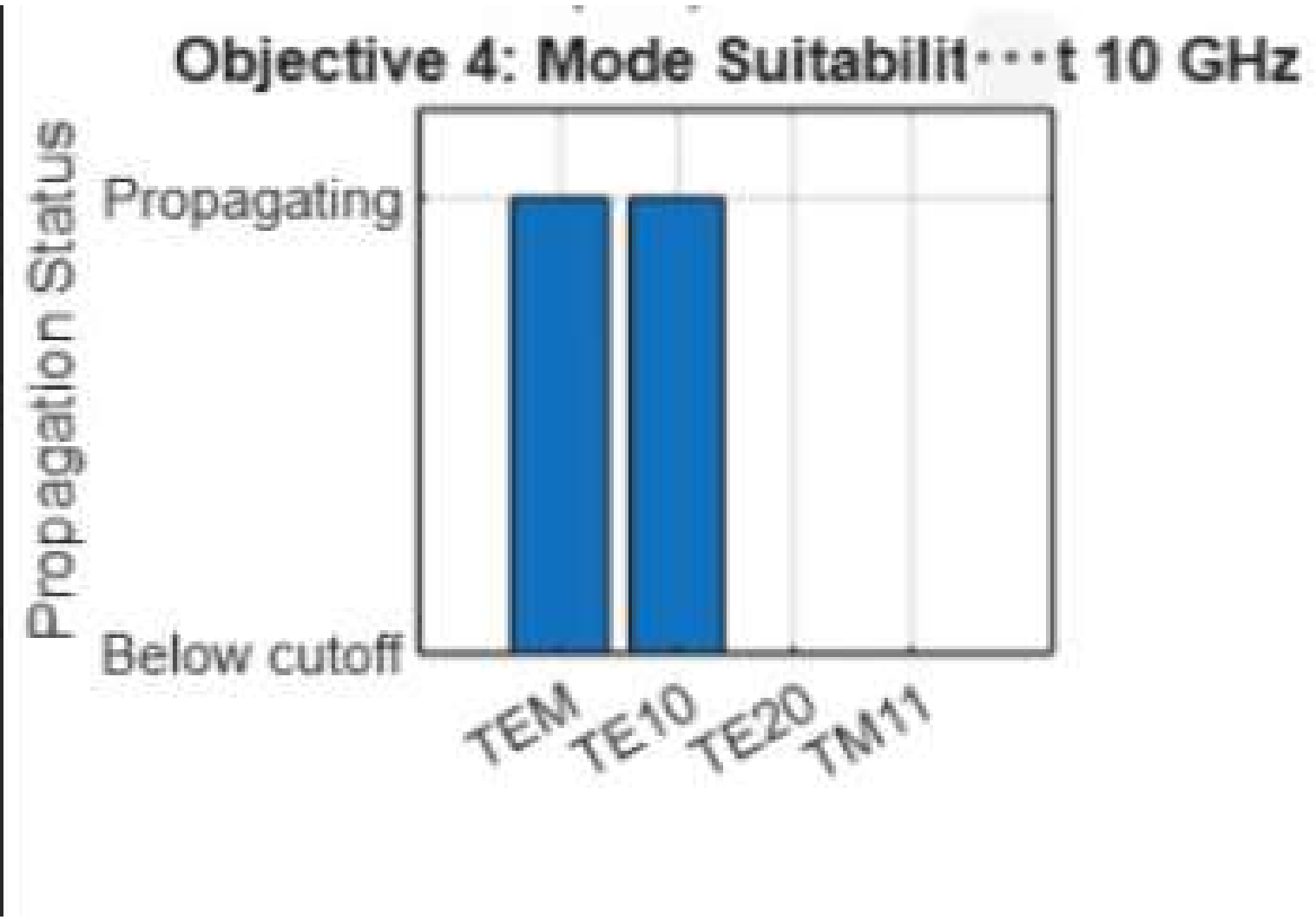

Objective 1: [illegible] Mode
Normalized E-field
1
0.5
0
-10
0
10
10
0
-10
1
0.8
0.6
0.4
0.2
rary units)
Y (arbitrary units)


Objective 2: Electri[illegible]eld (TE10)
10
5
0
Y (mm)
0
10
20
X (mm)
0.8
0.6
0.4
0.2
0


Objective 2: Magnetic F[illegible]d (Illustrative)
10
5
0
Y (mm)
0
10
20
X (mm)
1
0.8
0.6
0.4
0.2


Objective 3: Propagation ···nstant
600
400
200
β (rad/m)
TEM
TE10
TE20
TM11
5
10
15
20
25
Frequency (GHz)


Objective 3: Cutoff Freq···ncies
Cutoff Frequency (GHz)
20
15
10
5
0
TEM
TE10
TE20
TM11


Objective 4: Mode Suitabilit···t 10 GHz
Propagation Status
Propagating
Below cutoff
TEM
TE10
TE20
TM11

```
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 MICROWAVE ENGINEERING - EXPERIMENT 1
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Operating Frequency = 10.00 GHz

CUTOFF FREQUENCIES
TEM  = 0 GHz (ideal two-conductor line)
TE10 = 7.500 GHz
TE20 = 15.000 GHz
TM11 = 16.771 GHz

PROPAGATION CONSTANTS AT 10.00 GHz
TEM  beta = 209.440 rad/m
TE10 beta = 138.531 rad/m
TE20 is below cutoff at 10.00 GHz
TM11 is below cutoff at 10.00 GHz

PROPAGATION STATUS AT 10.00 GHz
TE10: Propagating
TE20: Below cutoff
TM11: Below cutoff
```